Blade pitch and propeller thrust

Analysis of data.zip | Measurements dated 3 November 2026

Main result: 15° has the highest recorded mean burst maximum, 0.5110 N, with 20° close behind at 0.5080 N. The difference is only 0.0030 N (0.59%) and is not robust to the measured zero offsets. Treat 15°-20° as the leading tested region for this motor, supply and 10-second full-throttle protocol.

Different question, different apparent optimum: exploratory estimates at 4,500-4,800 rpm favor roughly 40°-45°. These are transient-inclusive estimates, not controlled steady-RPM tests.

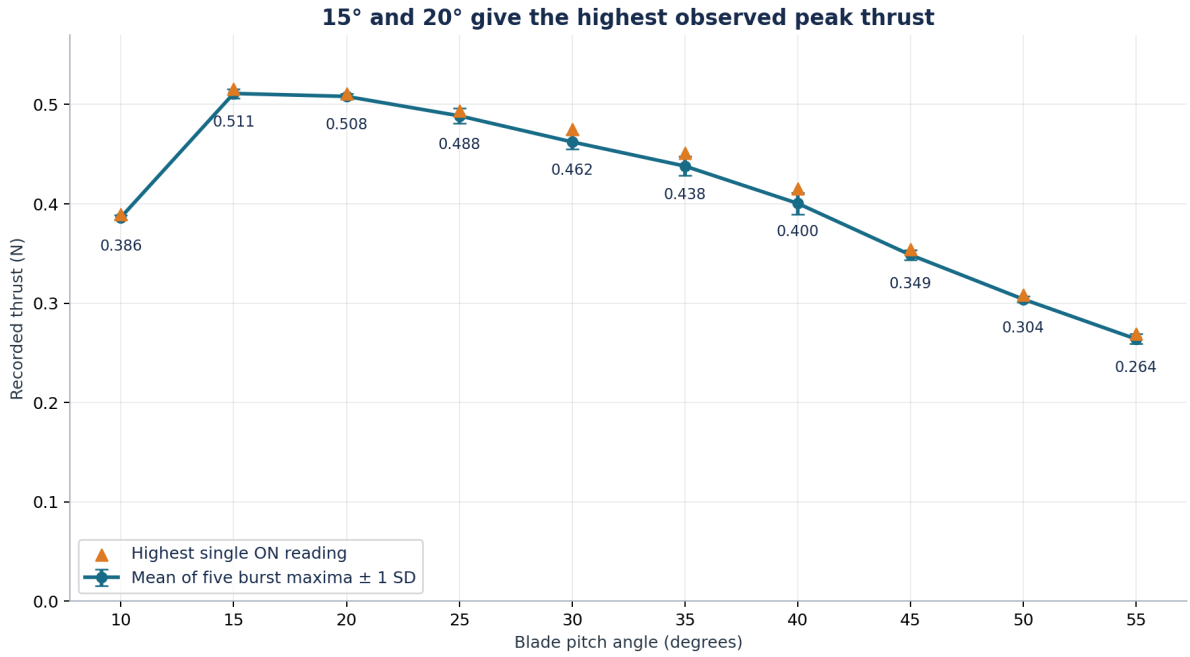
Pitch	Highest ON thrust (N)	RPM at that reading	Mean of 5 maxima ± sample SD (N)	Mean RPM at 5 maxima
10°	0.3894	11,700	0.3863 ± 0.0022	11,940
15°	0.5158	10,500	0.5110 ± 0.0046	10,380
20°	0.5109	9,000	0.5080 ± 0.0028	9,180
25°	0.4938	8,100	0.4885 ± 0.0076	8,100
30°	0.4750	7,200	0.4622 ± 0.0074	7,140
35°	0.4511	6,600	0.4379 ± 0.0097	6,660
40°	0.4151	6,300	0.4004 ± 0.0110	6,240
45°	0.3541	5,700	0.3485 ± 0.0052	5,760
50°	0.3085	5,400	0.3039 ± 0.0027	5,400
55°	0.2693	4,800	0.2639 ± 0.0050	4,860

What is being averaged

For each of the five powered bursts, take the largest signed **thrust_N** reading. The average maximum is the arithmetic mean of those five maxima: **mean peak = (Tmax,1 + ... + Tmax,5) / 5**. It is not the average of every sample in a run. The RPM in column 3 is from the same CSV row as the largest ON thrust, not the highest RPM elsewhere in the run.

Error bars and ± values use the sample standard deviation of five burst results (denominator 4). They show repeatability only, not total measurement uncertainty or a confidence interval. Five sequential bursts from one recorded trial are not five independently rebuilt or remounted experiments. All files use prop_id = 1 and trial = 1; they cannot establish variation between physical propeller specimens.

Peak thrust across the tested angles



100% throttle, 10-second bursts. SD describes burst repeatability; calibration uncertainty is not included.

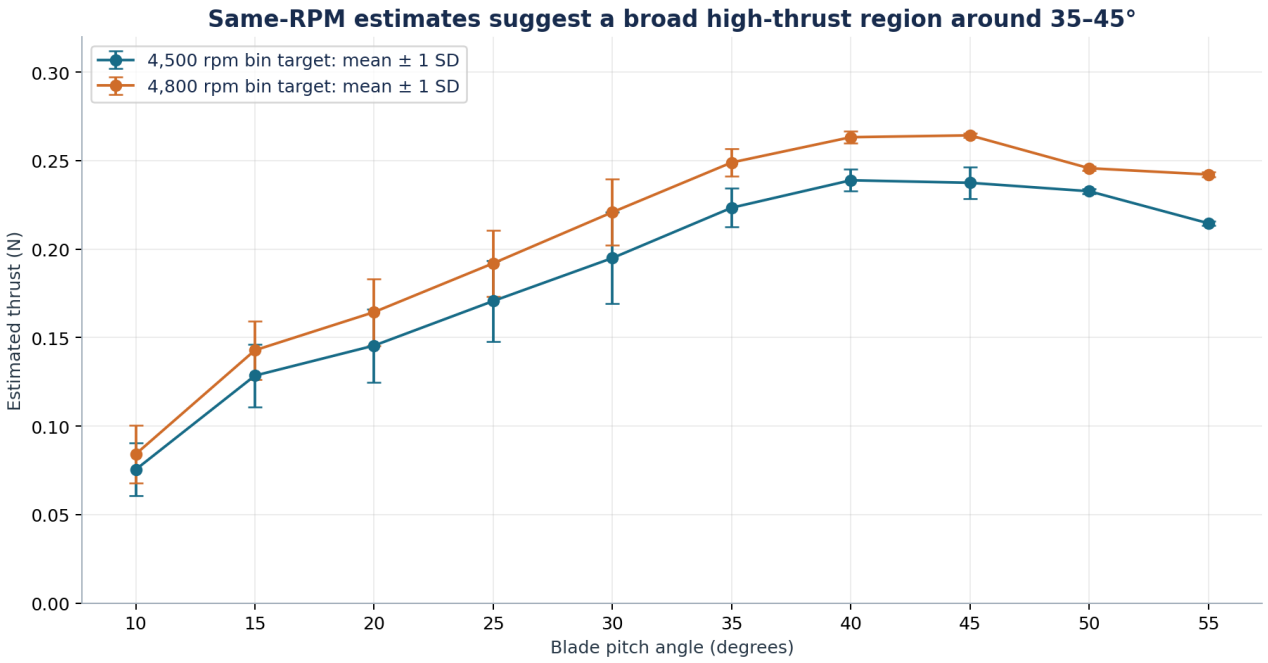
The mean maximum rises by **32.3%** from 10° to 15°, is nearly unchanged at 20°, and then declines to 55°. At 55°, it is **48.4% below** the 15° result. Repeatability SD ranges from 0.55% to 2.76% of the mean; 40° has the greatest relative burst-to-burst scatter.

Each burst: maximum thrust (N) @ recorded RPM

Pitch	Burst 1	Burst 2	Burst 3	Burst 4	Burst 5
10°	0.3832 @ 12,000	0.3894 @ 11,700	0.3860 @ 12,000	0.3865 @ 12,000	0.3865 @ 12,000
15°	0.5041 @ 10,200	0.5146 @ 10,500	0.5158 @ 10,500	0.5107 @ 10,200	0.5098 @ 10,500
20°	0.5082 @ 9,300	0.5034 @ 9,300	0.5109 @ 9,000	0.5090 @ 9,300	0.5086 @ 9,000
25°	0.4908 @ 8,100	0.4938 @ 8,100	0.4755 @ 8,100	0.4937 @ 8,100	0.4886 @ 8,100
30°	0.4587 @ 7,200	0.4608 @ 7,200	0.4557 @ 7,200	0.4611 @ 6,900	0.4750 @ 7,200
35°	0.4262 @ 6,600	0.4342 @ 6,600	0.4511 @ 6,600	0.4442 @ 6,900	0.4339 @ 6,600
40°	0.3883 @ 6,000	0.3907 @ 6,300	0.4151 @ 6,300	0.4019 @ 6,300	0.4058 @ 6,300
45°	0.3541 @ 5,700	0.3534 @ 5,700	0.3484 @ 5,700	0.3437 @ 6,000	0.3431 @ 5,700
50°	0.3011 @ 5,400	0.3036 @ 5,400	0.3029 @ 5,400	0.3036 @ 5,400	0.3085 @ 5,400
55°	0.2693 @ 4,800	0.2691 @ 5,100	0.2597 @ 4,800	0.2591 @ 4,800	0.2622 @ 4,800

Maxima use powered samples only, consistent with the supplied firmware summaries. Some first post-shutoff samples are slightly higher because the propeller and measurement system do not respond instantaneously; record-wide maxima are listed on page 8. No raw readings were deleted or silently corrected.

Comparing all pitches at lower RPM



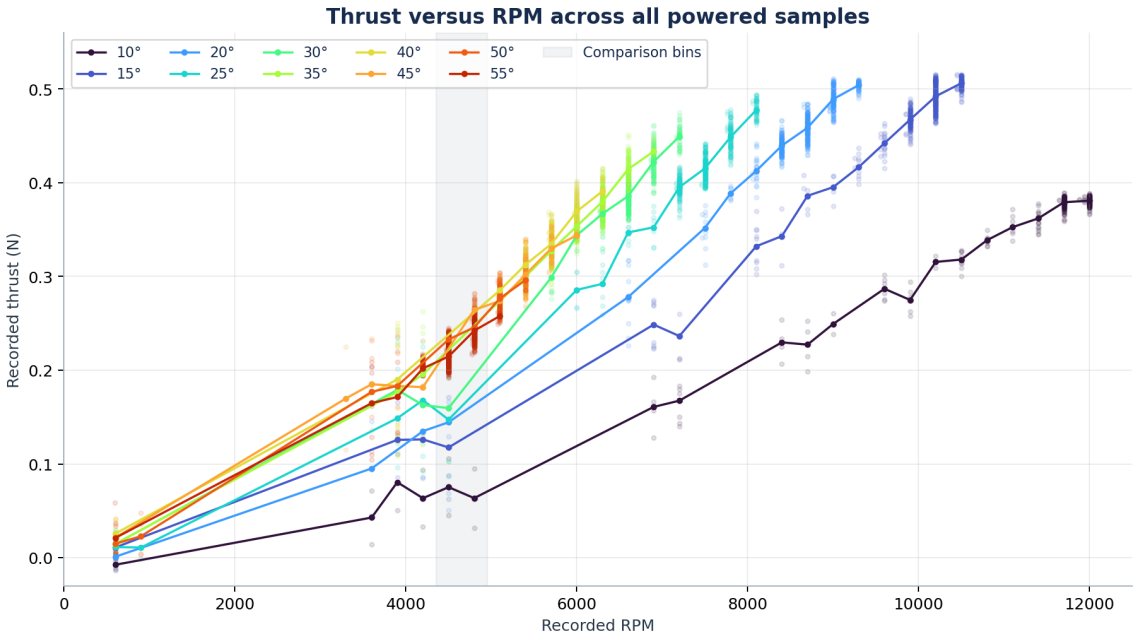
Exploratory only: 300-rpm bins; interpolation where missing; acceleration and later ON samples mixed.

Pitch	4,500 rpm estimate mean ± SD (N)	4,800 rpm estimate mean ± SD (N)	Bursts with observed bin at 4,500 / 4,800 rpm
10°	0.075 ± 0.015	0.084 ± 0.016	1/5 / 1/5
15°	0.129 ± 0.018	0.143 ± 0.017	3/5 / 0/5
20°	0.146 ± 0.021	0.164 ± 0.019	3/5 / 0/5
25°	0.171 ± 0.023	0.192 ± 0.019	2/5 / 0/5
30°	0.195 ± 0.026	0.221 ± 0.019	1/5 / 0/5
35°	0.223 ± 0.011	0.249 ± 0.008	0/5 / 0/5
40°	0.239 ± 0.006	0.263 ± 0.003	0/5 / 0/5
45°	0.238 ± 0.009	0.264 ± 0.001	0/5 / 5/5
50°	0.233 ± 0.001	0.246 ± 0.001	5/5 / 5/5
55°	0.215 ± 0.001	0.242 ± 0.001	5/5 / 5/5

Interpretation: the nominal highest estimates are 40° at 4,500 rpm (0.239 N) and 45° at 4,800 rpm (0.264 N). However, 40° and 45° are very close, and the interpolation and timing errors are not included in these SDs. This does not establish a precise optimal angle. These are estimated values at bin centers, not measured exact-RPM setpoints.

All five bursts contribute to each displayed mean. Where the bin is absent, interpolation supplies the estimate. In particular, neither target bin is directly observed in any 35° or 40° burst. Small error bars there do not imply small interpolation uncertainty.

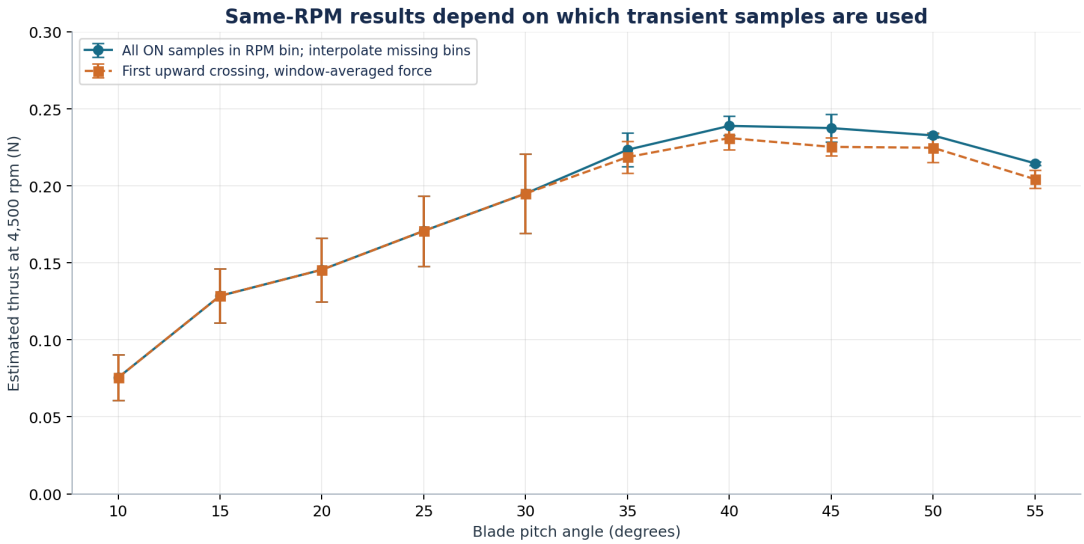
How the RPM comparison was made



Dots: valid-RPM ON samples. Lines: descriptive bin means, not steady-state aerodynamic fits.

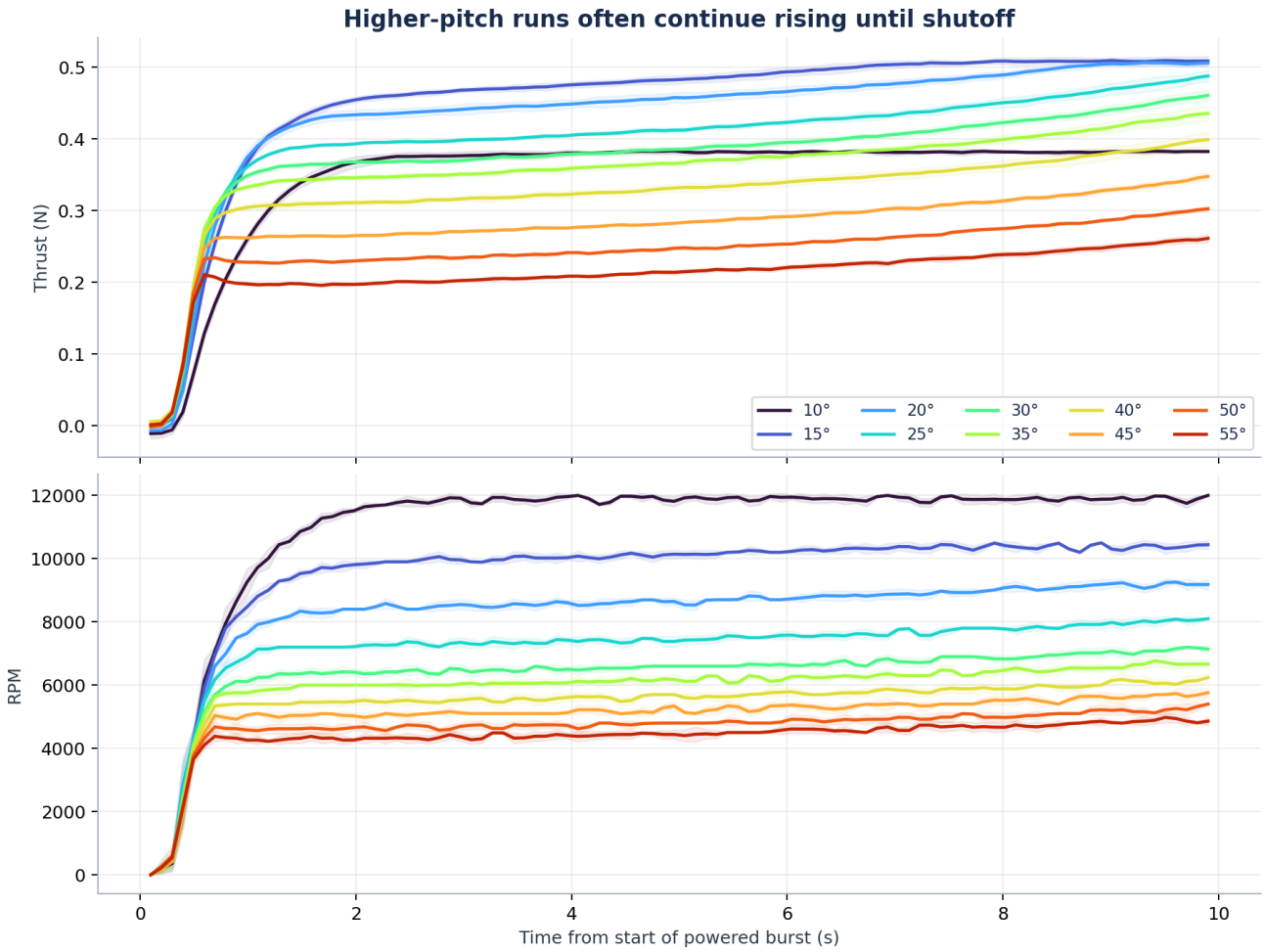
Only ON samples with rpm_valid = 1 and positive RPM are used. RPM is grouped into bins centered every 300 rpm (nominal width ±150 rpm); mean thrust is calculated separately for each burst and bin. Missing target bins are linearly interpolated between occupied bins in that same burst. No extrapolation is used. The five burst estimates then receive equal weight.

The targets 4,500 and 4,800 rpm lie inside the measured positive-RPM span of all 50 bursts. This is not continuous observational coverage: the largest interpolation bracket spans **4,200 rpm** at the 4,500 target, and **3,300 rpm** at the 4,800 target, both for 10°. Therefore these estimates are low-confidence comparisons, especially for low pitch.



As a sensitivity check, force readings sharing each RPM measurement window were averaged and only the first upward target crossing was interpolated. At 4,800 rpm this changes 40° from 0.263 to 0.251 N and 45° from 0.264 to 0.247 N, reversing their order. The general high-thrust region remains, but the exact winner is method-dependent.

Time dependence limits the peak claim



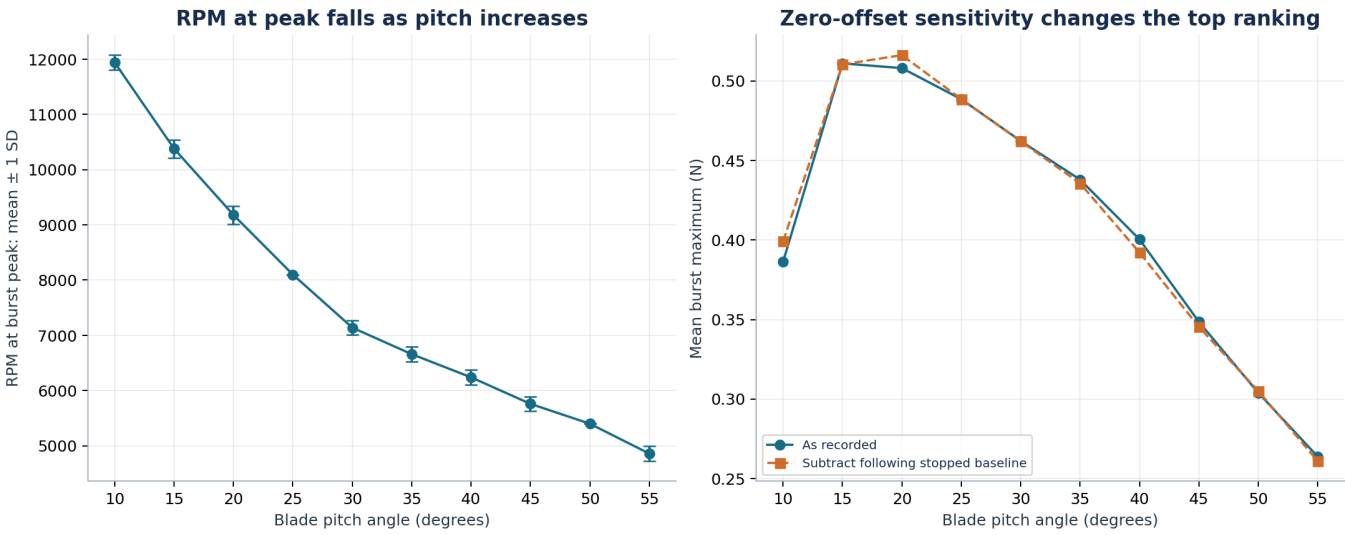
Five bursts aligned by elapsed time; lines are means and shading is ± 1 SD across bursts.

All 35 bursts at pitches 25°-55° reach their largest ON reading after 9.8 seconds, immediately before the commanded shutoff. Thus these are maxima observed within a fixed 10-second window, not demonstrated ultimate or equilibrium maxima.

Comparing the mean thrust at 8-10 seconds with 2-4 seconds, thrust rises by 1.5% at 10°, 9.0% at 15°, 13.8% at 20°, and 17.7%-23.2% at 25°-55°. RPM rises too. Most high-pitch curves have not settled by the end of the burst.

A longer run could change the apparent full-throttle ranking, but these data cannot predict how much or whether it would do so. The 5-10 second averages supplied in the CSV and on page 8 are descriptive late-window averages; they are deliberately not called steady-state thrust.

Zero offset is large enough to matter



Baseline adjustment is a sensitivity check, not a validated correction. Each burst uses its following 15–20 s OFF mean.

For each burst, the following 15-20 second segment is used to estimate the stopped baseline. Subtracting that mean from that burst's peak gives a sensitivity check. It assumes this stopped offset also applied while powered, which is not proven; the original values remain the primary reported measurements.

Pitch	Mean stopped baseline (N)	Recorded mean maximum (N)	Baseline-adjusted mean maximum (N)
10°	-0.0128	0.3863	0.3991
15°	+0.0007	0.5110	0.5103
20°	-0.0082	0.5080	0.5162
25°	-0.0001	0.4885	0.4886
30°	+0.0002	0.4622	0.4620
35°	+0.0023	0.4379	0.4356
40°	+0.0083	0.4004	0.3921
45°	+0.0031	0.3485	0.3454
50°	-0.0011	0.3039	0.3051
55°	+0.0028	0.2639	0.2611

The top ranking reverses: 15° becomes 0.5103 N and 20° becomes 0.5162 N. This is why the raw 0.0030 N advantage at 15° is not a defensible basis for claiming it is definitely better than 20°.

Investigate tare drift, load-cell creep or hysteresis, vibration, fixture loads and calibration. These are possible explanations, not diagnosed causes. A nonzero stopped baseline alone does not tell us the correct powered-state adjustment.

Physics interpretation and IA evaluation

Why the two comparisons differ

The full-throttle experiment changes pitch while allowing RPM to change. Average RPM at the burst maxima falls from 11,940 rpm at 10° to 4,860 rpm at 55°. Increased blade angle can increase aerodynamic torque demand and lower attainable speed for a given drive system [1]. This is consistent with the data, but voltage, current and torque were not recorded, so the cause cannot be isolated.

Propeller thrust is commonly represented as $\mathbf{T = C_T \rho n^2 D■}$, with n in revolutions per second, D the diameter and C_T a coefficient that depends on operating condition and geometry [2]. Both RPM and blade behavior matter. The coefficient cannot be assumed constant across different pitches, and normalizing every point by RPM² would not turn these transients into valid steady-RPM measurements.

The fall beyond the intermediate-pitch region at similar RPM could reflect increased drag or separated flow. It is not proof of stall. Geometric pitch angle is not the same as local aerodynamic angle of attack. Power efficiency also cannot be ranked from thrust and RPM alone; electrical input power or shaft torque is missing.

What to improve first

1. Match the research question to the measurement. For practical motor-system performance, use maximum or settled thrust at fixed throttle and supply conditions. For the effect of pitch at fixed RPM, actively hold a shared RPM setpoint. These are different experiments.

2. Collect actual steady-RPM data. A first shared target near 4,000 rpm is below the observed powered capability of every pitch; establish it experimentally, hold until thrust and RPM stop trending, then average a predeclared interval. Add further achievable setpoints. A target near 4,800 rpm leaves little margin for 55°.

3. Verify zero and calibration. Record stopped baselines before and after each run, calibrate with known loads over the measurement range and document force uncertainty. Verify the optical sensor really produces one accepted pulse per revolution. The logs use roughly 200 ms count windows: one pulse corresponds to about 300 rpm, so fine RPM precision is unsupported.

4. Control confounders and repeats. Record supply voltage and current, monitor heating, keep diameter/blade shape/angle reference consistent, and randomize pitch order. Repeat full trials with independent re-zeroing and remounting. The current sequence is 20°, 25°, 30°, 10°, 15°, 35°, 40°, 45°, 50°, 55°; time, battery state and temperature could be confounded with pitch.

A defensible conclusion

Under the tested 100%-throttle, 10-second protocol, the largest observed thrust occurred in the 15°-20° region. Exploratory comparisons near 4,500-4,800 rpm suggest a higher-pitch region around 40°-45° may give greater thrust at the same rotational speed. However, transient sampling, coarse RPM resolution, interpolation and baseline offsets prevent determination of a precise fixed-RPM optimum.

[1] FAA, Aviation Maintenance Technician Handbook - Powerplant, propeller principles. FAA handbook (PDF).
[2] NASA-hosted propeller system-identification paper, thrust coefficient definition. NASA technical paper (PDF). Accessed 3 November 2026. Numerical findings in this report are calculated from the supplied data, not these background sources.

Data audit and additional summaries

Ten raw CSVs contain 10,600 rows: 5,300 ON and 5,300 OFF, with 106 ON samples per burst. All runs are marked COMPLETE. Of the ON samples, 142 have invalid RPM and are excluded from RPM comparisons; they are retained for force calculations. All 50 ON peak rows have valid RPM. Recomputed mean maxima match all ten supplied summary JSON files to within 1e-9 N.

Largest reading anywhere in the recorded run

Pitch	Maximum (N)	Phase	RPM in same row	Burst / elapsed time (s)
10°	0.3894	on	11,700	2 / 4.973
15°	0.5158	on	10,500	3 / 9.023
20°	0.5113	off	9,300	4 / 10.037
25°	0.4938	on	8,100	2 / 9.970
30°	0.4780	off	7,200	5 / 10.035
35°	0.4578	off	6,900	3 / 10.062
40°	0.4151	on	6,300	3 / 9.971
45°	0.3569	off	6,000	1 / 10.080
50°	0.3094	off	5,400	5 / 10.049
55°	0.2693	on	4,800	1 / 9.987

Late-window means: 5-10 seconds of each powered burst

Pitch	Mean thrust ± burst SD (N)	Mean RPM
10°	0.3817 ± 0.0018	11,899
15°	0.5017 ± 0.0049	10,322
20°	0.4838 ± 0.0037	8,943
25°	0.4453 ± 0.0055	7,739
30°	0.4174 ± 0.0047	6,838
35°	0.3960 ± 0.0057	6,398
40°	0.3596 ± 0.0069	5,876
45°	0.3104 ± 0.0022	5,443
50°	0.2708 ± 0.0009	5,010
55°	0.2351 ± 0.0033	4,676

The accompanying package includes six figures in PNG and SVG, pitch and burst summary CSVs, matched-RPM estimates and per-burst interpolation brackets, per-burst RPM-bin data, combined raw data with cycle time, the original input CSV/metadata/summary JSON files, and Python scripts. Extra 6,000-rpm estimates are in the CSV but are not a ten-pitch comparison: 50° and 55° never reach it; only two 45° bursts do.